

I Taught a Computer to Read Brain Waves for Seizures.

One patient broke the pattern.

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Imagine this. It's late, and your screen is the only light in the room. In front of you are two monitors, side by side. Both of them are displaying recordings of the same exact brain within the span of an hour. Almost every single time, they would both read the same. However this case is a little different. At one point it's normal. At the other, it's in the middle of a seizure.

You don't need a computer to tell you which is which. You can see it with your own two eyes. When the brain is normal, the window moves calmly and steadily. On the other hand, when the seizure is active, the window is chaotic, almost like someone grabbed the line and shook it.

One is a heartbeat while the other is chaos.

Then you start thinking. If you are able to differentiate how a brain looks normally versus during a seizure, could a computer learn to see it too?

That question is the foundation of this article.

Imagine living with epilepsy. Not knowing when the next seizure hits. Not mid-conversation, not driving, not asleep, there's no warning system built into your own body telling you it's coming. One moment is completely normal. The next isn't.

Living with epilepsy is a condition that no average person would be able to fully imagine. To ask for someone to stay strong with a condition at this level is a big ask. But that's the purpose of this article. To understand if technology has evolved enough to a point where it can detect if you are mid-seizure or not based on the brain state.

The point of this article isn't to research on how epilepsy can be cured. No. It's one small, honest question, buried underneath a much bigger one everyone likes to jump straight to.

I searched for hours but no matter what, I couldn't find a credible answer to this question. So I decided to find it myself.

I got myself real EEG recordings, the same kind of data hospitals actually use. It's essentially an hour of one person's brain, recorded in immense detail. There are 23 different “sensing” devices

across the scalp, each with the rate of sampling at 256 Hz.

We need to ask ourselves, how do you turn “the seizure line looks messier” into something a computer can do? This is where the concept of variance comes in. It's a math term, but it simply just measures how much something swings up and down. A calm, steady line has a low variance while a wild, jumpy line has a high variance.

My first step was to cut the whole recording into thousands of tiny two-second windows. For every single one, I calculate the variance. I then divided all of the windows into two categories: seizure and non-seizure.

Then came the most important part. Comparing the two piles.

The gap wasn't subtle in any way. The seizure windows averaged about nine times higher variance than a normal window. This number isn't something controversial. It's a number big enough that you could build a simple rule around it and rarely be wrong.

However, in reality, that was just one case. That result could just be luck, so I ran it again. This time on a completely new seizure, but from the same person.

Followed the same process. Got very similar results. About 8.3 times higher.

Two for two. For a second, I actually thought I'd cracked it without much effort.

But the next level from here is to test this same procedure on a completely new patient to see if this pattern still applies.

I got the windows. Calculated the variance. Separated them into groups. Calculated the results expecting the same thing.

It barely worked at all.

Instead of that dramatic nine times difference, this patient's seizure variance was only about 1.3 times higher than normal. Almost nothing. If I hadn't already known exactly where the seizure was hiding in that hour of data, I never would have found it using this number alone.

My initial thought was that something was broken. But this made me realize as someone conducts research, a result you didn't expect isn't automatically a failure, it's just a clue that you haven't figured everything out yet. Something about this second brain was genuinely different, and the method I'd trusted twice in a row had just told me nothing.

I stared at the data, the procedure, to see why the results were different. And then it hit me. I was only looking at one single spot on the brain. The same exact channel, every single time, because it happened to work so well on the first patient.

So then I began to wonder, what if the problem was never the method. What if I was just looking in the wrong place the whole time.

I had 23 channels given to me, but to simplify the research initially, I had only analyzed one. Now I used all 23.

And there it was. The channel I'd been using this whole time, the one that worked so well on the first patient, turned out to be one of the weakest spots on this brain. But other channels told a completely different story. One channel, P8-O2, showed a variance 7.59 times higher during the seizure. Another, F3-C3, came in at 7.15 times. A third, T8-P8, at 5.32 times.

The seizure had been there the entire time. I just hadn't been looking in the right place to see it.

There was one more twist though. Two of the channels, C4-P4 and P4-O2, actually showed the opposite pattern, the normal activity had higher variance than the seizure did. I still don't fully know why. But it told me something important, this isn't as simple as finding the one right channel and using it forever. Every brain seems to hide its signal somewhere different.

The fix was never a smarter formula. It was just utilizing my resources.

This whole article is backed around one small question. Could a computer tell the difference between a calm brain and a seizing one?

The honest answer is yes, but only if you're willing to look in more than one place to find it. What worked perfectly for one person, didn't for the next. The fix wasn't some brilliant new formula, it was simply opening my mind and realizing that the location could be the reason for the different results.

However, that's only a small discovery in the grand scheme of things. It's not a seizure-prediction device, it's not a cure, it's not even close to something a hospital could use tomorrow. But it's an honest answer, backed by real research, on a question that had to be answered before anyone could take the next step.

Somewhere out there, someone is still working on the bigger version of this question, whether a computer could catch the warning signs minutes before a seizure even starts. I don't have the knowledge and authority to answer that specific question.

But this is what I do know. A little more about what it actually takes to answer that bigger question than I did a month ago, and that's what truly matters the most.

And to end off, if there is one thing this whole article taught me, through the research, documentation, and writing; it's that the answer is rarely sitting in the first place you look. Sometimes you just have to check all 23.

REFERENCES

Guttag, John. "CHB-MIT Scalp EEG Database" (version 1.0.0). *PhysioNet* (2010). RRID:SCR_007345.
<https://doi.org/10.13026/C2K01R>

Goldberger, A. L., et al. "PhysioBank, PhysioToolkit, and PhysioNet: Components of a New Research Resource for Complex Physiologic Signals." *Circulation*, vol. 101, no. 23, 13 June 2000, pp. e215-e220, 10.1161/01.CIR.101.23.e215.

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